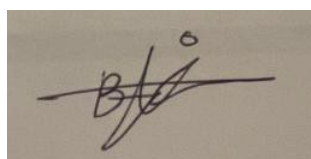


COMM061 NLP Coursework Report

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Course Name	NLP(Natural Language Processing)
Date Submitted	05 / 05 / 26

[Bong jun kim]



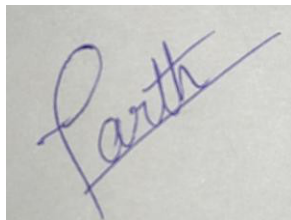
[Piyush Gupta]
Signature:



[Akash Chohan]



[Parth Singh Jhala]



[Turab Ghumman]



1. Analyse and visualise the dataset

1.1 Visualising Distributions and Domain Specificity

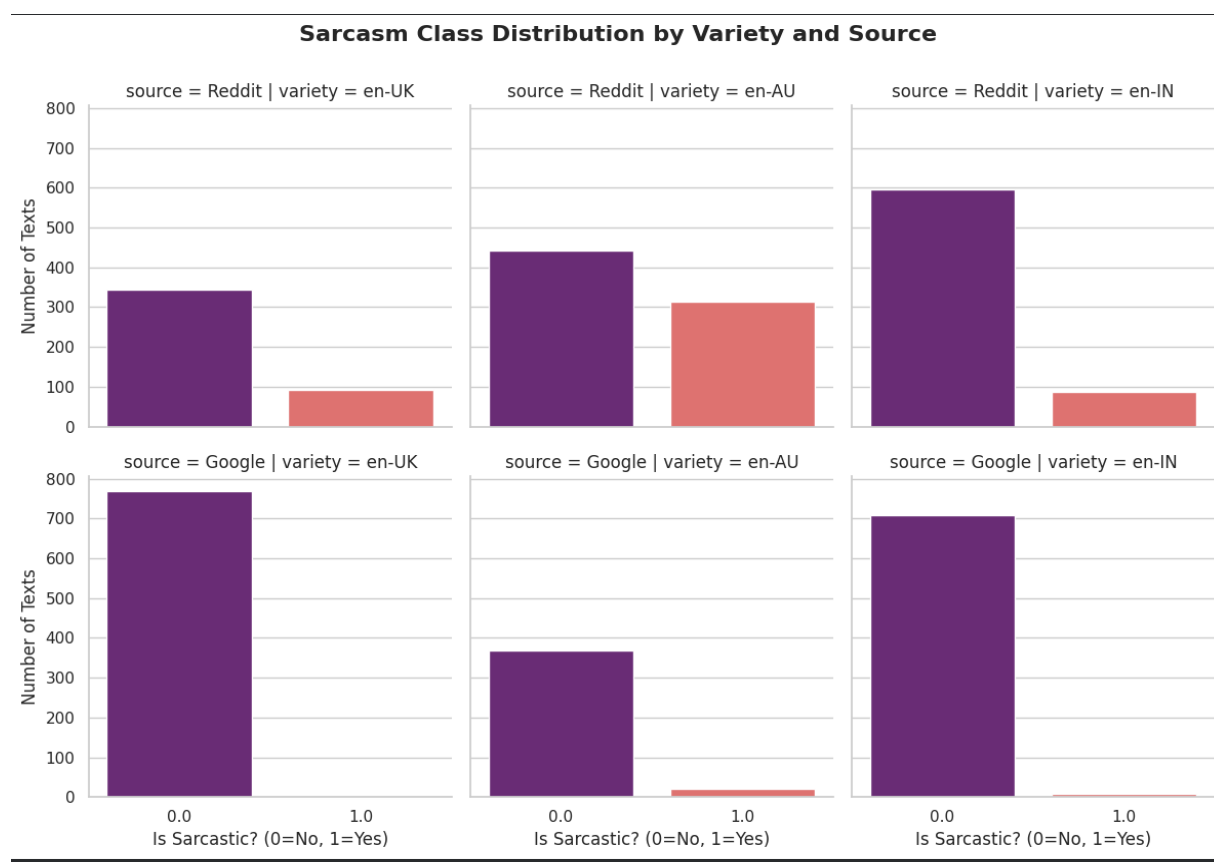
As part of the initial exploration of the data, we faceted the original dataset based on the English dialects present (en-UK, en-AU, and en-IN) and the source platform (Google Places and Reddit). This faceting of the original dataset based on the source platform resulted in a massive disparity between the two platforms. To begin with, the dataset sourced from Google Places is dominated by positive sentiment with a significant absence of sarcasm present in the reviews.



[Figure 1. Sentiment distribution by variety and source.]

This could be due to the professional nature of the platform. On the other hand, the Reddit dataset has a high number of negative sentiment and sarcasm-dominant reviews, with the disparity being more significant in the case of the Australian (en-AU) subset of the dataset. Given the severity of the class imbalance of the Sarcasm label in the real-world dataset, the team has quickly recognised the fact that 'Accuracy' would be a highly misleading evaluation criterion in this scenario. This is because a model could simply always choose 'Not Sarcastic' and still obtain a high accuracy score.

Therefore, we will rely on the Macro F1-score as our primary evaluation metric to ensure both classes are weighted equally.



[Figure 2. Sarcasm distribution by variety and source.]

1.2 Vocabulary Analysis and Linguistic Markers

To accurately pinpoint the true linguistic markers among the dialects, we opted to use the TF-IDF (Term Frequency-Inverse Document Frequency) Vectorizer, as opposed to the word count method. The math behind the TF-IDF Vectorizer effectively penalizes extremely common words, such as "the" or "and," and instead helps us pinpoint the most unique slang and dialect words available in each region. However, the most important part is that this analysis was done only on the training set. If the analysis was performed on the test set, this would essentially give the model "prior knowledge" of the test set's features, thus defeating the entire purpose and scientific nature of the test. The analysis was successful, and the model was able to pinpoint the unique region-specific markers. The markers found for the Australian (en-AU) dialect included "mate," "reckon," "bloody," "straya," and "no worries." The markers found for the British (en-UK) dialect included "brilliant," "cheers," "innit," "proper," and "rubbish." The markers found for the Indian (en-IN) dialect included "prepone," "revert," "kindly," "dear," and "needful."

1.3 The Pre-processing Strategy

The data cleaning process followed in this data cleaning pipeline was a mix of standard string operations and sophisticated filtering techniques. Lowercasing all the text was done to ensure uniformity among all tokens. We have utilized regular expressions (regex) to remove noisy HTML tags and URLs, which were prevalent in the Reddit data set. Unlike traditional data preprocessing pipelines for NLP tasks, where all non-alphanumeric characters are removed, we have chosen to preserve all punctuation and emojis. Our team is working on fine-tuned variants of Transformer-based models like RoBERTa for a Sarcasm data set. Removing punctuation like "..." or "!!!!" would essentially destroy all syntactical cues and emotional nuances for a model to recognize sarcasm.

2. Experimentation with three different setups

2.1 Baseline, Transformer (PTLM Gap)

2.1.1 Experimental Setup

In this experiment, we compare the performance of a classical machine learning baseline with a fine-tuned pretrained transformer model on both sentiment classification and sarcasm detection tasks. The objective is to quantify the performance gap between traditional feature-based methods and modern contextual language models.

The baseline model consists of a TF-IDF vectorizer combined with Logistic Regression. The TF-IDF representation uses unigram and bigram features to capture lexical patterns within the text. This approach is widely used in traditional NLP pipelines due to its simplicity and efficiency, but it is limited in its ability to capture contextual and semantic relationships.

In contrast, we fine-tune a RoBERTa-base model, a transformer-based architecture pretrained on large-scale corpora. RoBERTa leverages contextual embeddings, allowing it to capture deeper semantic and syntactic relationships within the text. The model is fine-tuned separately for sentiment and sarcasm classification tasks using the same dataset splits as the baseline model, ensuring a fair comparison.

Both models are trained on the same training set and evaluated on the same test set. Performance is measured using accuracy, precision, recall, and Macro-F1 score, with Macro-F1 serving as the primary evaluation metric due to its robustness in imbalanced classification settings.

Table 1 summarises the experimental pipeline used for the classical baseline and the transformer-based model. The key difference is that the baseline relies on sparse lexical TF-IDF features, while RoBERTa uses contextual token representations learned during pretraining and adapted through fine-tuning.

Step	TF+IDF + Logistic Regression	RoBERTa
1	Load cleaned train/validation/test splits	Load cleaned train/validation/test splits
2	Convert text into TF-IDF unigram/bigram features	Tokenize text using RoBERTa tokenizer
3	Train Logistic Regression classifier	Fine-tune RoBERTa for sequence classification
4	Predict labels on the test set	Predict labels on the test set
5	Evaluate using Accuracy, Precision, Recall, Macro-F1	Evaluate using Accuracy, Precision, Recall, Macro-F1

[Table 1. Experimental pipeline for baseline and RoBERTa models]

2.1.2 Results

The performance of both models across the sentiment and sarcasm tasks is summarised in Table 2.

Task	Model	Accuracy	Macro-Precision	Macro-Recall	Macro-F1
Sentiment	TF-IDF+LR	0.8246	0.8255	0.8239	0.8241
Sentiment	RoBERTa	0.8914	0.8922	0.8920	0.8914
Sarcasm	TF-IDF+LR	0.7604	0.6132	0.6809	0.6249
Sarcasm	RoBERTa	0.8704	0.7295	0.6459	0.6730

[Table 2 Performance comparison between TF-IDF and RoBERTa]

2.1.3 Analysis

The results demonstrate a clear performance gap between the baseline and transformer-based approaches, particularly for sentiment classification. RoBERTa achieves a Macro-F1 score of 0.8914, significantly outperforming the TF-IDF baseline, which achieves 0.8241. This improvement can be attributed to RoBERTa's ability to model contextual relationships and capture nuanced semantic information, which is essential for understanding sentiment. For sarcasm detection, the performance improvement is notably smaller. RoBERTa achieves a Macro-F1 score of 0.6730, compared to 0.6249 for the baseline. While this still represents an improvement, the gain is relatively modest compared to the sentiment task. This suggests that sarcasm detection is inherently more challenging and cannot be fully addressed by simply increasing model complexity.

A key observation is that sarcasm relies heavily on implicit meaning, pragmatic context, and cultural cues, which are not always explicitly represented in the text. As a result, even advanced transformer models struggle to generalise effectively in this task. Furthermore, the gap between accuracy and Macro-F1 in sarcasm classification indicates potential class imbalance effects. While accuracy remains relatively high, the lower Macro-F1 score suggests that the model may still be biased toward the majority class, reinforcing the importance of using Macro-F1 as the primary evaluation metric.

2.2 Cross-variety Experimentation

2.2.1 Experimental Setup

This experiment evaluates how well models trained on one English variety generalise to others. Unlike Section 2.1, each variety (en-AU, en-IN, en-UK) is used as an isolated training source. Models are trained on one variety and tested on the remaining two, resulting in six transfer directions per task. No cross-variety data is included during training, ensuring a strict zero-shot setting.

The baseline uses TF-IDF with unigram and bigram features (1–2 n-grams), sublinear TF scaling, a vocabulary cap of 20,000, and Logistic Regression with balanced class weights. RoBERTa-base is fine-tuned using the same configuration as Section 2.1 (max length 128, batch size 16, learning rate 2e-5, up to 3 epochs with early stopping).

All models are evaluated using Macro-F1 as the primary metric due to class imbalance. Evaluation is conducted on the same test set as Section 2.1, with results reported per target variety

2.2.2 Results

Train	Test	TF-IDF F1	RoBERTa F1
en-AU	en-IN	0.4952	0.5110
en-AU	en-UK	0.5937	0.6092
en-IN	en-AU	0.4475	0.4139
en-IN	en-UK	0.4950	0.4803
en-UK	en-AU	0.4782	0.4139
en-UK	en-IN	0.5614	0.4822

[Table 3. Cross-Variety Sarcasm Detection – Macro F1]

Train	Test	TF-IDF F1	RoBERTa F1
en-AU	en-IN	0.7543	0.8137
en-AU	en-UK	0.8559	0.9399

Train	Test	TF-IDF F1	RoBERTa F1
en-IN	en-AU	0.7696	0.8763
en-IN	en-UK	0.8512	0.9298
en-UK	en-AU	0.7672	0.7871
en-UK	en-IN	0.7649	0.8210

[Table 4. Cross-Variety Sentiment Analysis – Macro F1]

Tables 3 and 4 present cross-variety Macro-F1 scores for sarcasm detection and sentiment classification.

For sentiment classification, both models achieve consistently strong performance across all transfer directions. In contrast, sarcasm detection shows substantially lower and more variable performance, with most transfer pairs yielding Macro-F1 scores around 0.41–0.50. Models trained on en-AU achieve relatively higher performance compared to those trained on en-IN or en-UK.

2.2.3 Analysis

The results indicate a clear difference between the two tasks. Sentiment classification transfers effectively across all varieties, whereas sarcasm detection shows limited generalisation.

For sarcasm detection, models trained on en-IN or en-UK exhibit strong bias toward the majority class, suggesting that the training data does not provide sufficient signal for learning the minority class. In contrast, models trained on en-AU achieve better transfer performance, indicating that training data composition plays a critical role.

2.2.4 TF-IDF vs RoBERTa

For sentiment classification, RoBERTa consistently outperforms TF-IDF across all transfer pairs.

For sarcasm detection, both models show similarly limited performance under most transfer conditions, suggesting that differences in model architecture are less influential than the characteristics of the training data.

2.2.5 Interpretability Analysis(LIME and SHAP)

Interpretability analysis was conducted on the worst-performing transfer pair (en-IN → en-AU, sarcasm).

LIME shows that predictions are strongly influenced by high-frequency function words (e.g., “of”, “the”, “in”), while potential sarcasm indicators have limited impact. SHAP confirms that many influential features are high-frequency function words, alongside some content words.

Both analyses indicate that the model relies on shallow lexical signals rather than transferable patterns, and that predictions are biased toward the majority class.

2.3 LLM Variations

2.3.1 Overviews

In this section, we apply the fine-tuning approach of Low-Rank Adaptation (LoRA) based on the pre trained frozen Llama 3.2 1B model to detect sarcasm in two dialects of English: en-UK and en-IN. We train two adapters separately for each dialect and conduct a cross-evaluation of the two adapters on the test data for the two dialects using a 2×2 matrix. The reason why we use LoRA fine-tuning instead of regular fine-tuning is because it allows us to save the number of trainable parameters from ~1B to ~4–6M per adapter, which can be efficiently trained on the free tier of Colab T4 GPUs, and also the adapter swapping method fits our deployment purpose in Section 5.1.

2.3.2 Dataset and Class Imbalance

The BESSTIE Dataset (surrey-nlp/BESSTIE-CW-26) offers predetermined train/validation/test splits for each variety. The two varieties used in this section are characterized by heavy sarcasm class imbalance, which directly affects the performance measures.

Variety	Training Samples	Sarcasm Rate	Neg:Pos Ratio	Mitigation
En-UK	~2,000+	-7.7%	-12:1	Oversample positives to 1:1
En-IN	~2,000+	-6.8%	-14:1	Oversample positives to 1:1

[Table 5. Class imbalance in training data and mitigation strategy]

Why accuracy is misleading: A classifier which predicts "Not Sarcastic" for every example yields a near perfect accuracy of 92–93%, but it fails to detect any sarcasm. That's why we use Macro-F1 throughout the project – because it treats both categories equally, irrespective of their frequencies. The difference between high accuracy (around 90-93%) and low Macro-F1 isn't an error – it's exactly what you should expect when dealing with imbalanced data. This is similar to the findings in Section 2.2, where he obtains an accuracy of 91-93% along with Macro-F1 ranging from 0.41 to 0.50. The minority category ("Sarcastic") is randomly oversampled to match the majority category ("Not Sarcastic") by a ratio of 1:1 for each variety within the training loop.

2.3.3 Model Architecture and LoRA Configuration

The base model is meta-llama/Llama-3.2-1B, Quantised in 4-bit NF4 through BitsAndBytes from ~4 GB (full precision) to ~1 GB of GPU memory. The LoRA adapter has been designed using PEFT and trained via SFTTrainer from TRL.

Instruction format: "you are an assistant that can detect sarcasm. Is the text below sarcastic? Yes or no only please." This makes use of the instruction-following nature and the real-world understanding of sarcasm by Llama, without training a classification head on it.

Hyperparameter	Value	Justification
Base model	Llama 3.2 1B	Open-weight; within required 1B–3B range; strong instruction-following
Quantisation	4-bit NF4	~1 GB VRAM; enables Colab T4 training within resource constraints
Rank(r)	16	r=8 risks underfitting on binary task; r=32 adds cost with no measurable gain
LoRA_alpha	32	2×r — standard LoRA initialisation (Hu et al., 2021) for unit-scale update
LoRA_dropout	0.05	Light regularisation preventing over-reliance on single low-rank directions
Target modules	q/k/v/o/gate/up/down	All 7 attention + feed-forward projections for richer representations
Learning rate	2e-4	Standard LoRA LR; higher than full fine-tuning as only adapter matrices update
Epochs	3	Converges without overfitting; best checkpoint by validation loss (eval_strategy='epoch')
Precision	Bf16	Wider dynamic range than fp16; reduces gradient underflow with 4-bit quantisation
Trainable params	~4–6M	vs ~1B in full model; base weights remain completely frozen throughout

[Table 6. LoRA hyperparameter and justification]

2.3.4 Training Procedure

Two adapters are trained successively, with the same set of hyperparameters, except for the variety subset. Each adapter's training cycle consists of the following operations:

- Data preparation: Training data is limited to the desired variety; oversampling of minority classes to 1:1 ratio; keeping validation data with natural class distribution.
- Model preparation: Loading Llama 3.2 1B from scratch in 4-bit NF4 mode and setting `device_map='auto'`, inserting LoRA adapter through `get_peft_model()`
- Training: Running SFTTrainer for 3 epochs with cosine learning rate scheduler, 50 warmup steps, gradient accumulation steps=4 (effective batch size=16), bf16 floating-point format.
- Checkpoint selection: Selecting best checkpoint on the basis of minimum validation loss using `load_best_model_at_end=True`; removing base model after training to prevent uploading huge files.
- Saving: Saving adapter weights to `./adapter_en-uk` and `./adapter_en_in`; while deploying, loading it through `PeftModel.from_pretrained()` based on the common frozen base.
- Multiple training runs: The adapter was trained twice using different random seeds to check for consistency the most effective checkpoint.

2.3.5 Evaluation Design

Prediction method — logits, not generation: Instead of calling `model.generate()` and then parsing the text output (which can be unreliable and prone to case sensitivity issues), the model takes a direct look at the logits output from the forward pass and compares the log probabilities of the yes and no tokens. This is mathematically sound and will not be affected by generation artifacts.

Threshold tuning: Since Llama's probability distribution is learned from pre-training data and not from the sarcasm classification problem, a static 0.5 threshold is not ideal. Threshold tuning from 0.10 to 0.90 in steps of 0.02 finds the optimal cutoff value that maximizes Macro-F1 score. This is mainly how sarcasm examples that are misclassified as Not Sarcastic when using a 0.5 threshold can be recovered.

Adapter trained on	Tested on en-UK	Tested on en-IN
En-UK adapter	In-variety (res_uk_uk)	Cross-variety (res_uk_in)
En-IN adapter	Cross-variety (res_in_uk)	In-variety (res_in_in)

[Table 7. 2X2 cross-evaluation matrix design]

3. Evaluation

3.1 TF-IDF + Logistic Regression, RoBERTa

This experiment compares a traditional TF-IDF + Logistic Regression baseline with a fine-tuned RoBERTa model for both sentiment classification and sarcasm detection. The evaluation focuses on Macro-F1 as the primary metric, along with accuracy, macro precision, and macro recall.

3.1.1 Overall performance comparison

Task	Model	Accuracy	Macro-Precision	Macro-Recall	Macro-F1
Sentiment	TF-IDF+LR	0.8246	0.8255	0.8239	0.8241
Sentiment	RoBERTa	0.8942	0.8947	0.8937	0.8940
Sarcasm	TF-IDF+LR	0.7604	0.6132	0.6809	0.6249
Sarcasm	RoBERTa	0.8690	0.7244	0.6475	0.6814

[Table 8. Overall performance comparison]

RoBERTa significantly outperforms the baseline in both tasks. The improvement is substantial for sentiment classification, where Macro-F1 increases from 0.8241 to 0.8940.

For sarcasm detection, the improvement is more modest (0.6249 to 0.6814), indicating that sarcasm is inherently more challenging due to its reliance on implicit and contextual cues.

3.1.2 Per-Class Performance

Task	Class	Precision	Recall	F1-score	Support
Sentiment	Negative	0.8845	0.9123	0.8982	1117
Sentiment	Positive	0.9049	0.8752	0.8898	1066
Sarcasm	Non-sarcasm	0.9020	0.9510	0.9259	1878
Sarcasm	Sarcasm	0.5468	0.3639	0.4370	305

[Table 9. Per-Class Performance]

For sentiment classification, performance is well balanced across both classes.

In contrast, sarcasm detection shows a large performance gap. While the non-sarcasm class achieves a high F1-score (0.9259), the sarcasm class performs poorly (0.4370), primarily due to low recall. This indicates that the model fails to detect many sarcastic instances.

3.1.3 Majority-Class Bias

Task	Class	True Count	Predicted Count
Sentiment	Negative	1117	1152
Sentiment	Positive	1066	1031
Sarcasm	Non-sarcasm	1878	1980
Sarcasm	Sarcasm	305	203

[Table 10. Majority-Class Bias]

The results show that for sarcasm detection, the model heavily favours the non-sarcasm class. This bias leads to relatively high accuracy but poor recall for sarcasm. This highlights why Macro-F1 is a more appropriate metric than accuracy in imbalanced classification tasks.

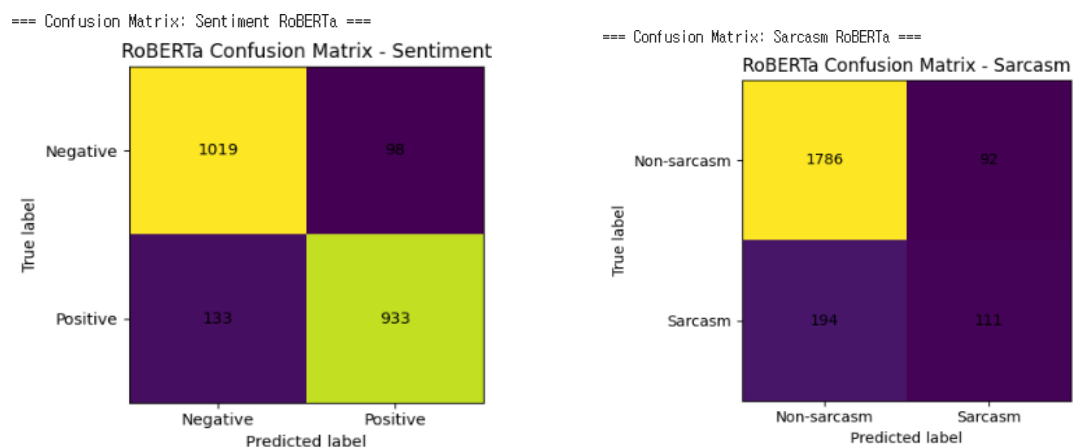
3.1.4 Cross-Variety Performance

Task	Variety	Accuracy	Macro F1
Sentiment	en-AU	0.8966	0.8957
Sentiment	en-IN	0.8529	0.8524
Sentiment	en-UK	0.9400	0.9399
Sarcasm	en-AU	0.7826	0.6918
Sarcasm	en-IN	0.9093	0.5249
Sarcasm	en-UK	0.9043	0.6969

[Table 11. Performance across English varieties(RoBERTa)]

Sentiment classification is robust across varieties. However, sarcasm detection varies significantly, with particularly low performance on en-IN. This suggests that sarcasm expressions differ across dialects, making generalisation difficult.

3.1.5 Error Analysis



[Figure 3. Sentiment / Sarcasm confusion matrix]

The sentiment confusion matrix shows balanced misclassification between classes, indicating no strong bias. In contrast, the sarcasm confusion matrix reveals that many sarcastic instances are misclassified as non-sarcastic. This confirms that the model lacks sensitivity to sarcasm.

3.2 Cross-variety Evaluation

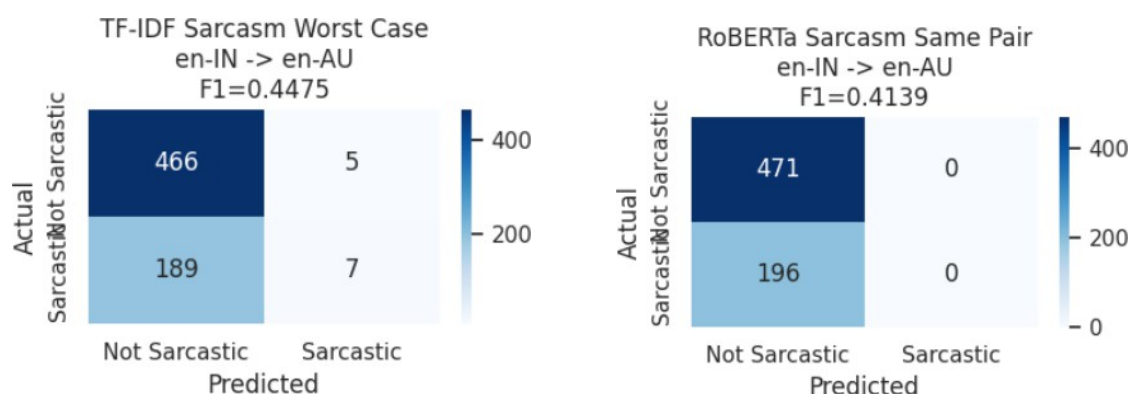
We evaluate cross-variety transferability using RoBERTa-base, fine-tuned on each English variety (en-AU, en-IN, en-UK) and tested on the others. As shown in Table 3 and Table 4, performance varies depending on the source-target pair. For example, en-AU $\rightarrow$ en-UK achieves relatively strong performance (macro-F1 = 0.6284), whereas en-IN $\rightarrow$ en-AU performs poorly (macro-F1 = 0.4139). This pattern suggests that transferability is influenced by linguistic similarity between varieties.

The results reveal a clear asymmetry between the two tasks: sentiment classification transfers well across all variety pairs, whereas sarcasm detection exhibits significantly weaker and more unstable performance.

3.2.1 Sarcasm Detection: Worst-Case Analysis

The worst-performing transfer pair for sarcasm detection is en-IN $\rightarrow$ en-AU. The confusion matrix (Figure 4) shows that the model predicts almost all instances as “Not Sarcastic,” with all sarcastic instances misclassified, resulting in zero recall for the sarcastic class. This corresponds to near-majority-class collapse, indicating that the model fails to learn a meaningful decision boundary for the sarcastic class.

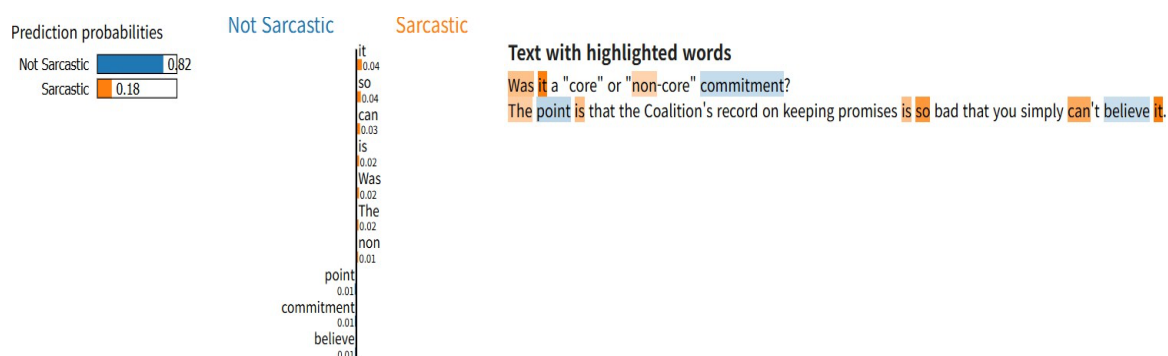
To further analyse this behaviour, LIME is applied to misclassified instances. The explanations reveal that the model relies heavily on common tokens such as “the”, “of”, and “is”, while failing to capture meaningful sarcasm cues. This suggests that the model fails to capture deeper contextual and pragmatic signals required for identifying sarcasm across different language varieties.



[Figure 4 Confusion Matrix for TF-IDF and RoBERTa(en-IN → en-AU)]

3.2.2 LIME-Based Error Analysis

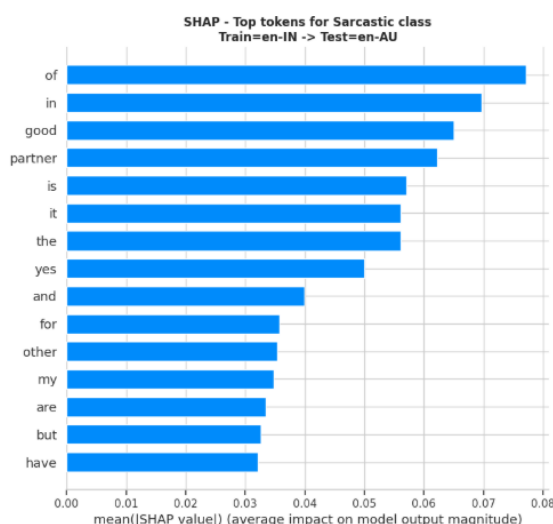
To analyse misclassification behaviour, LIME is applied to incorrectly predicted instances. The results show that the model assigns high importance to common tokens such as “the”, “of”, and “is”, which are not informative for sarcasm detection. In contrast, contextually meaningful expressions are often ignored, leading to incorrect predictions. This indicates that the model relies on shallow lexical features rather than capturing the contextual and pragmatic cues required for identifying sarcasm.



[Figure 5 LIME Explanation for Misclassified Instance]

3.2.3 SHAP-Based Error Analysis

SHAP is used to analyse feature importance at a global level. As shown in Figure 6, the most influential features are high-frequency tokens, confirming that the model's predictions are dominated by shallow lexical patterns. This finding is consistent with the LIME analysis and further supports the conclusion that the model fails to capture meaningful contextual signals for sarcasm detection.



[Figure 6 SHAP summary plot]

3.2.4 Sentiment vs Sarcasm

Unlike sarcasm detection, sentiment classification remains stable across all transfer settings. As shown in Table 4, both TF-IDF and RoBERTa achieve consistently high macro-F1 scores, with only minor variation across transfer pairs. This suggests that sentiment relies on generalisable linguistic features that transfer well across dialects.

In contrast, sarcasm detection exhibits large performance variation and frequent failure cases, often corresponding to near-majority-class collapse. This highlights its reliance on deeper contextual understanding and its sensitivity to dialectal and distributional differences.

3.3 LLM Variations

3.3.1 Overall Performance Comparison

Adapter	Tested on	Accuracy	Precision	Recall	Macro-F1	Threshold
En-UK	En-UK	0.7357	0.6113	0.8570	0.5987	0.88
En-UK	En-IN	0.5674	0.5663	0.7595	0.4678	0.88

En-IN	En-UK	0.7529	0.6128	0.8490	0.6086	0.88
En-IN	En-IN	0.5944	0.5682	0.7657	0.4843	0.88

[Table 12. Full results across all 4 adapter-variety combinations]

The LoRA-based adapters obtain decent Macro-F1 (0.47–0.61). Even though accuracy seems pretty high (~0.57–0.75), it does not provide a reliable measurement as accuracy is dependent on the class imbalance, see Section A.2. As expected, in-variety results (diagonal cells) are higher than cross-variety results (off-diagonal)

3.3.2 Per-Class Performance

Class	Precision	Recall	F1-score	Support
Not Sarcastic	1.00	0.74	0.85	647
Sarcastic	0.23	0.96	0.37	53

[Table 13. Per-Class Performance: en-IN adapter tested on en-UK]

As one could have guessed from the confusion matrix, the primary issue lies with the Sarcastic class. While non-sarcasm obtains very high precision (~1.0) but fairly average recall, Sarcastic class has the reverse situation, with very high recall (~0.96) but very poor precision (~0.23). This means that the model predicts that many sentences are sarcastic, even though they may be not (high sensitivity), resulting in a significant number of false positives. This is precisely what threshold tuning aims for, with the reduction in threshold allowing capturing more sarcasm cases, albeit with some errors in prediction. Threshold tuning is the optimal strategy for optimizing the Macro-F1 score since both False Negative and False Positive have an equal weight in the macro average.

3.3.3 Cross-Variety Performance

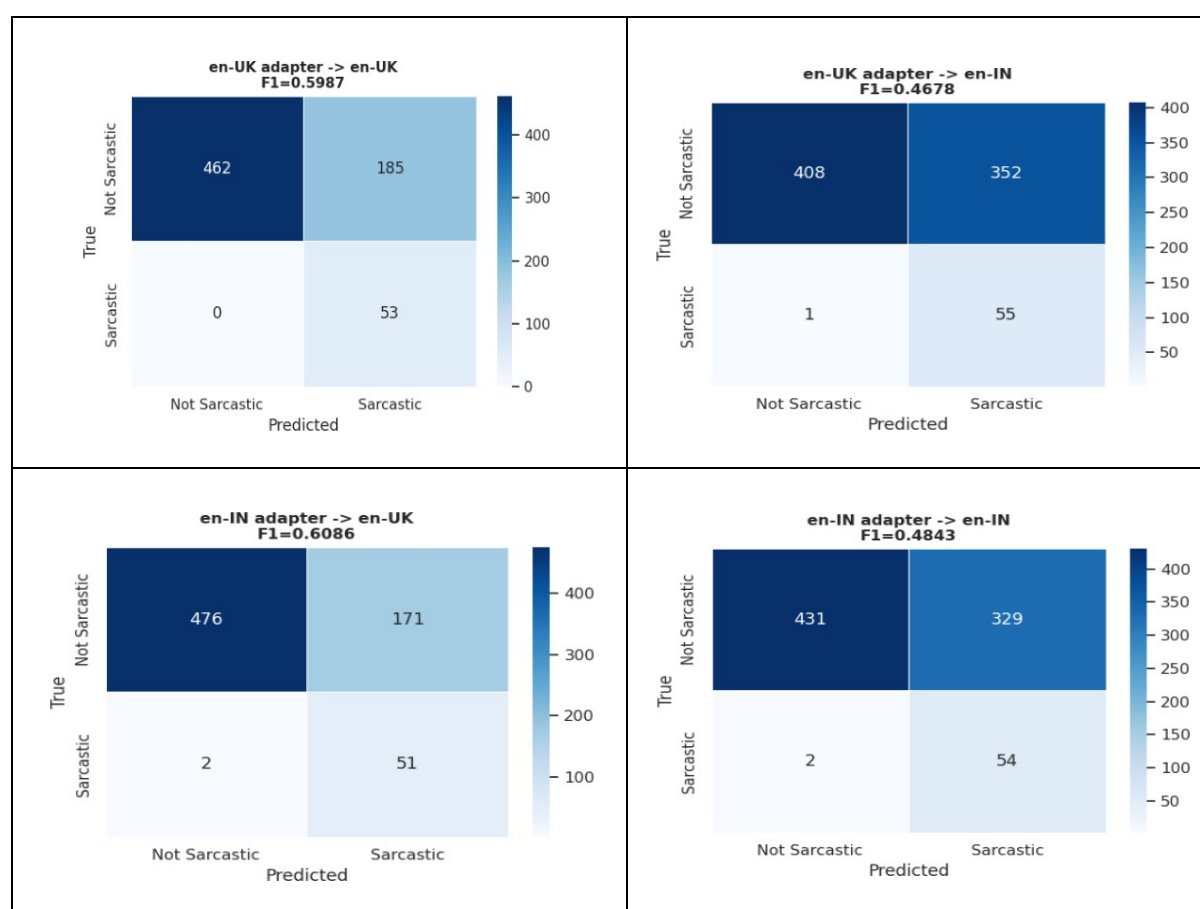
Adapter	In-variety F1	Cross-variety F1	Drop	Interpretation
En-UK	0.5987(on en-UK)	0.4678(on en-IN)	-0.1309	In-domain better; normal overfitting to UK cues
En-IN	0.4843(on en-IN)	0.6086(on en-UK)	+0.1243	Worse in-domain; en-UK test set may be easier

[Table 14. Cross-Variety Comparison]

The en-UK adapter demonstrates the expected behaviour with the higher in-variety Macro-F1 score (0.5987) and 0.4678 cross-variety, resulting in a difference of 0.13. This can be attributed to the adapter overfitting to surface features of British English, specifically to their tone, indirect nature of criticism, and lexical items such as 'brilliant' or 'cheers', which fail to generalize to other dialects.

On the contrary, en-IN adapter demonstrates the opposite trend with its lower in-variety Macro-F1 score (0.4843) as compared to the cross-variety (0.6086). It should be noted that it is neither the proof of better generalization nor overgeneralization. It just means that in addition to learning difficulties of the task itself, in-domain training data differs from testing data in terms of dataset difficulty. Namely, the en-IN adapter finds it difficult to categorize code-switched sentences with embedded rhetorical statements into sarcastic categories, as opposed to the more conventional sarcastic expressions in the en-UK test set.

3.3.4 Confusion Matrix Analysis



[Figure 7. Confusion matrix]

There are four confusion matrices (lora_confusion_matrices.png), each for one adapter-varieties combination. It is seen that the same pattern repeats itself in all cases:

- True Negatives are prevailing: Since there is the dominating non-sarcastic class, the models are right to classify non-sarcastic instances
- A good True Positive ratio: By adjusting the threshold, the model identifies most sarcastic instances, so recall is around 0.85 – 0.95 for the Sarcastic class
- A lot of False Positives: Due to lowering the threshold, there are numerous non-sarcastic instances classified as sarcastic, thus resulting on low precision (~0.23)
- There is almost no False Negative: The small number of missing sarcastic instances is expected for the metric of Macro-F1

The difference between accuracy and Macro-F1 is explained through these findings: the True Negatives make the accuracy score high, while a great number of False Positives leads to low precision.

3.4 Comparison to Section 2.2(RoBERTa)

RoBERTa — Llama 3.2 leverages the pragmatically trained understanding of sarcasm and irony obtained in its instruction-tuning process, complementing the small fine-tuning dataset. Nevertheless, LoRA Macro F1 scores (0.47–0.61) remain similar to, and not much superior to, RoBERTa's cross-variety scores (~0.41–0.50) in Section 2.2. This proves the core thesis of this coursework: architecture and model capacity do not matter as much as the training data quality in detecting sarcasm. Representational power is not the limiting factor here — whether RoBERTa or a 1B-parameter LLM, it struggles to detect sarcasm due to an inherent lack of sarcastic examples in the en-UK and en-IN training corpora.

4. Sarcasm Explanation & Error Analysis

4.1 Erroneous Predictions

To analyse model behaviour, 10 erroneous predictions are extracted from the best-performing LoRA adapter (en-IN → en-UK). Most selected errors correspond to false positives, where non-sarcastic instances are incorrectly classified as sarcastic, with only a small number of false negatives observed.

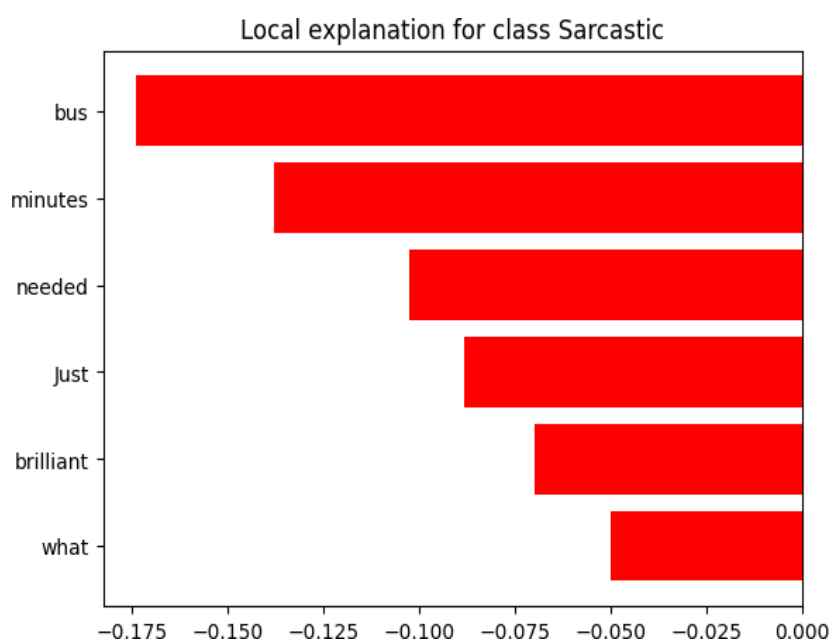
This indicates that the model exhibits a strong bias toward predicting sarcasm rather than making random errors. This pattern is consistent with the confusion matrix results, where the number of false positives is significantly higher than false negatives.

To further understand this behaviour, the false positive errors are analysed qualitatively. Most of the misclassified examples fall into the following categories:

From the qualitative analysis of these errors, four consistent types of mistakes emerge in classifying examples incorrectly:

- Statements of opinion/criticism: Non-ironic statements that express strong personal sentiments and opinions are flagged as sarcastic because the model associates negative sentiment with sarcasm.
- Rhetorical statements: Arguments or explanations, which may include factual information, are mislabelled as sarcastic because they appear confident and strong, similar to sarcastic expressions.
- Emotional expressions: Intensive language or expressions, including exclamation marks (“!!!!”), are detected as sarcastic because they are frequent in sarcastic texts but also occur in sincere emotional texts.
- Conversational statements: Phrases that sound casual and informal are marked sarcastic even though the intention is non-ironic, because the model associates informal registers with sarcasm.

These findings suggest that the model relies primarily on relatively shallow lexical or stylistic cues (e.g., tone, sentiment, and formality), rather than deeper contextual or pragmatic signals such as the inconsistency between literal meaning and intended interpretation. This conclusion aligns with the results from LIME analyses, where the RoBERTa model focused more on high-frequency function words and contextual cues.



[Figure 8. LIME feature importance for a sarcastic prediction using the fine-tuned RoBERTa model]

As demonstrated in the LIME visualization above, the RoBERTa model successfully captures the contextual nuance of sarcasm. Words typically associated with positive sentiment, such as “brilliant” and “Oh”, act as the strongest indicators for the “Sarcastic” class when placed in a negative real-world context (a delayed bus). Conversely, neutral logistical words pull slightly towards the “Not Sarcastic” class. This confirms that the RoBERTa encoder is interpreting semantic context, serving as a baseline comparison for our LLM errors.

4.2 Few-Shot prompt Construction

Text	Truth	Explanation (condensed)
The quality of the brit's~	Sarcastic	The sentence uses positive wording (“best”) to describe a clearly negative situation (poor road quality), creating a contradiction between surface meaning and actual intent.
Utter arsehole. ~	Not Sarcastic	The statement expresses a clear negative opinion (“utter arsehole”) directly, without any contrast between wording and intent, so it is not sarcastic.
It's actually insane to me that~	Not Sarcastic	The sentence expresses a strong opinion (“insane”) that directly reflects the speaker's belief, with no mismatch between surface wording and underlying meaning.
He turned number 10 into a ~	Not Sarcastic	The sentence describes a situation in a factual and literal manner, with no exaggeration or contrast that would indicate sarcasm.

[Table 15. Few-shot Examples with Explanations (see Appendix A)]

From the extracted error cases, four examples are selected and manually annotated with their ground-truth labels and explanations. These examples are used to construct a few-shot prompt to guide the model's reasoning. The explanations focus on the presence or absence of incongruity between literal meaning and intended interpretation, which is a key characteristic of sarcasm.

4.3 Few-Shot Evaluation

The remaining six erroneous examples are re-evaluated using the constructed few-shot prompt.

Text	Ground Truth	Original Prediction	Few-shot Prediction	Improved
These people are deranged extremists ~	Not Sarcastic	Sarcastic	Not Sarcastic	O
Not realistic. None of the players have gone ~	Not Sarcastic	Sarcastic	Not Sarcastic	O
The whole concept of “hate speech” is orwell ~	Sarcastic	Not Sarcastic	Not Sarcastic	X
Before all the inevitable “but China! posts ~	Not Sarcastic	Sarcastic	Not Sarcastic	O
Those poor children – no chance with a family ~	Not Sarcastic	Sarcastic	Not Sarcastic	O
That headline completely misrepresents the ~	Not Sarcastic	Sarcastic	Not Sarcastic	O

[Table 16. Few-shot Evaluation Results]

The remaining six erroneous examples are re-evaluated using the constructed few-shot prompt. The results show that 5 out of 6 cases are corrected after applying the few-shot prompt (83% improvement).

All improved cases correspond to false positives, where non-sarcastic texts are initially misclassified as sarcastic but are corrected after applying the prompt. The remaining false-negative case was not corrected, suggesting that few-shot prompting is more effective at reducing over-prediction of sarcasm than recovering missed sarcastic instances.

4.4 Final Summary

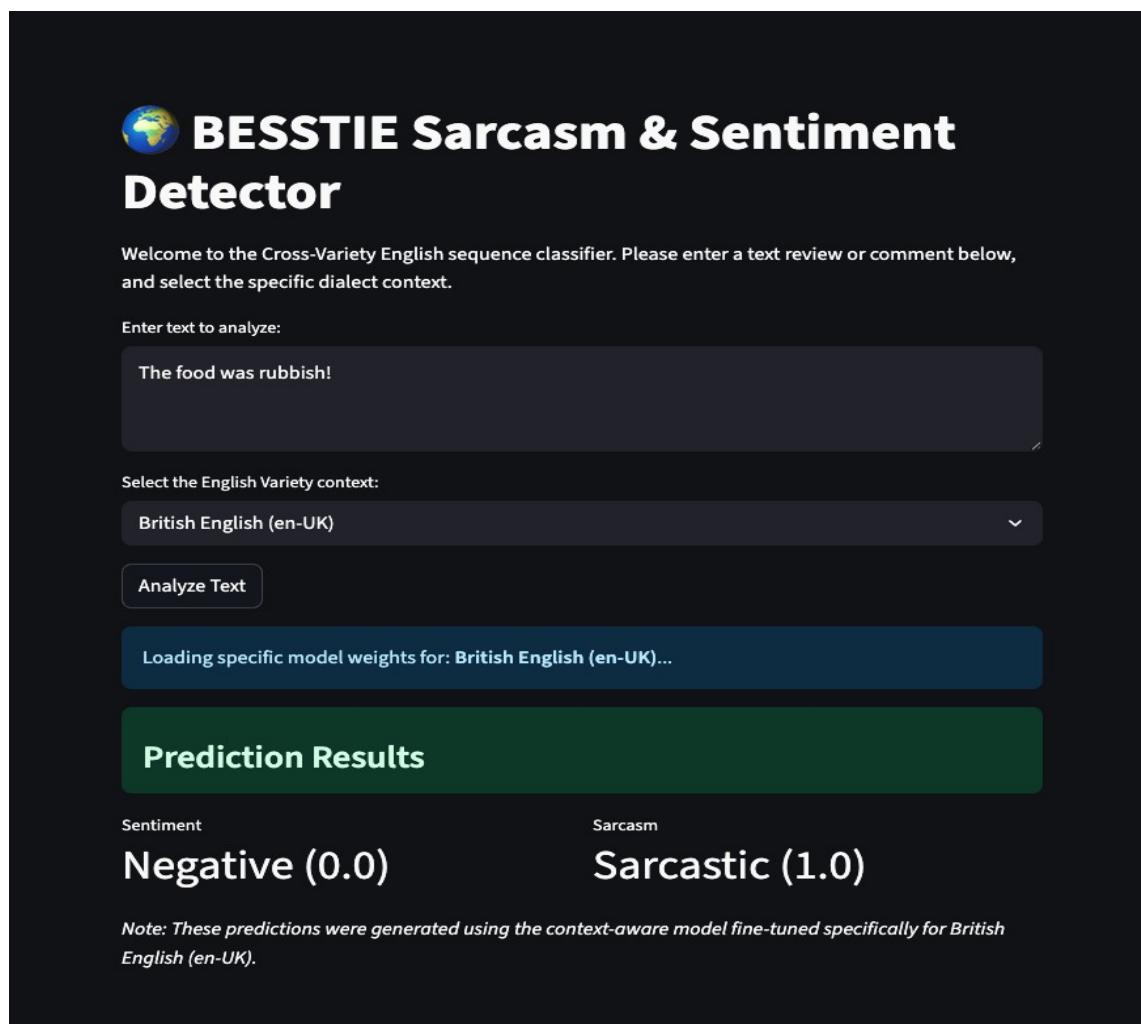
The results demonstrate that the few-shot prompt led to a clear improvement in erroneous predictions, but the effect was not uniform across error types. As shown in Table 16, 5 out of 6 selected cases were corrected after applying the revised 4-shot prompt. All five false-positive cases were successfully resolved, indicating that the model became more reliable in avoiding over-classification of sarcasm when no ironic contrast is present.

However, the few-shot prompt did not improve all erroneous predictions. One false-negative case remained misclassified, suggesting that the model still struggles to detect subtle or context-dependent sarcasm where the ironic intent is not explicitly expressed.

Overall, these findings indicate that few-shot prompting is effective in reducing false positives by reinforcing clear reasoning patterns but has limited capability in addressing false negatives that require deeper contextual understanding. This highlights a limitation of the current prompt-based approach, particularly in cases involving implicit or nuanced expressions.

5. Deployment

5.1 Deployment Endpoint



BESSTIE Sarcasm & Sentiment Detector

Welcome to the Cross-Variety English sequence classifier. Please enter a text review or comment below, and select the specific dialect context.

Enter text to analyze:

The food was rubbish!

Select the English Variety context:

British English (en-UK)

Analyze Text

Loading specific model weights for: British English (en-UK)...

Prediction Results

Sentiment	Sarcasm
Negative (0.0)	Sarcastic (1.0)

Note: These predictions were generated using the context-aware model fine-tuned specifically for British English (en-UK).

[Figure 9. Streamlit web interface demonstrating dialect-aware routing for Sarcasm and Sentiment detection.]

To deploy our best-performing models, we engineered a web application using the Streamlit framework. We selected Streamlit because it allows for rapid prototyping of data science applications entirely in Python, bridging the gap between our Jupyter notebooks and a user-facing product.

Crucially, the UI architecture requires the user to explicitly select the English variety (British, Australian, or Indian) via a dropdown menu before analysis. This allows the backend to dynamically route the input to the correct context-aware model. For example, if a user selects "British English," the system explicitly loads the model fine-tuned on the en-UK data split. Moving forward, this architecture is highly compatible with Parameter-Efficient Fine-Tuning (PEFT) methods like LoRA; rather than reloading an entire massive Transformer model into VRAM for every request, the application can simply hot-swap the tiny adapter weights specific to the chosen dialect, drastically reducing memory overhead and deployment costs.

5.2 Efficiency

To understand the trade-offs of deploying our models to a production environment, we benchmarked the inference latency of the classical TF-IDF + Logistic Regression baseline against the fine-tuned RoBERTa transformer. Testing was conducted on standard input (1 text) and a subsequent large batch (100 texts).

Model	Single Text Inference Time	Batch (100) Inference Time
TF-IDF + Logistic Regression	1.87 ms	11.04 ms
RoBERTa-base	83.41 ms	295.61 ms

[Table 17. Inference Latency Comparison (Baseline vs. RoBERTa)]

The results demonstrate a stark latency trade-off. While RoBERTa is vastly superior at detecting nuanced semantic meaning and context (resulting in much higher F1 scores for Sentiment), it is significantly slower—taking roughly 44x longer to process a single text compared to the baseline. Ultimately, the choice of model depends entirely on the deployment constraint. If we were building an offline pipeline for academic research, we would gladly sacrifice speed for RoBERTa's superior performance. However, if this system were deployed as a live content-moderation API handling thousands of requests per second, the baseline model's sub-2ms response time would be necessary to prevent bottlenecking the server architecture.

[Appendix A] Few-shot Prompt Construction Code

Instruction format: "you are an assistant that can detect sarcasm. Is the text below sarcastic? Yes or no only please."

Example 1: Text: The quality of the brit's Truth: Sarcastic
Explanation: The sentence uses positive wording ("best") to describe a clearly negative situation (poor road quality), creating a contradiction between surface meaning and actual intent.

Example 2: Text: Utter arsehole. Truth: Not Sarcastic
Explanation: The statement expresses a clear negative opinion ("utter arsehole") directly, without any contrast between wording and intent, so it is not sarcastic.

Example 3: Text: It's actually insane to me that Truth: Not Sarcastic
Explanation: The sentence expresses a strong opinion ("insane") that directly reflects the speaker's belief, with no mismatch between surface wording and underlying meaning.

Example 4: Text: He turned number 10 into a Truth: Not Sarcastic
Explanation: The sentence describes a situation in a factual and literal manner, with no exaggeration or contrast that would indicate sarcasm.

Now classify the following text: Text: {text}